

HONEYWELL EMBEDDED ENGINEERING TEAMS — PA · IA · BA

AI Champions Programme

Module 01

AI Champions Kick-off & Embedded Engineering Enterprise Context

Establishing the enterprise mandate, the AI Champion role, and the embedded engineering baseline — before GitHub Copilot, Spec-Driven Development, and Agent Prism enter the picture.



DURATION

1.5 Hours · Module 1 of 12



FORMAT

Workshop + Facilitated Discussion



DELIVERED FOR

Honeywell Embedded Engineering

How We'll Spend These Ninety Minutes

Six connected moves — from enterprise framing to a workshop output the whole programme builds on.



01

Kick-off, Why AI Champions & Survey Themes

10 min

Why Honeywell-style embedded teams need a champion model — plus what the pre-training calibration survey told us.



02

Pain-Point Workshop

20 min

Surface today's real delivery pain across PA, IA & BA embedded/firmware work.



03

AI Champion Role & Expectations

15 min

Champion / Evangelist / Sponsor personas mapped to Architects, Devs, Testers, DevOps, SRE, Leads.



04

Risks of Unmanaged AI Usage

15 min

Business & engineering risk of ungoverned agent use on closed-toolchain C/C++ firmware.



05

Lifecycle Context + Agent Prism

20 min

Where the embedded SDLC hurts today, and the monitoring lens we'll use all week.



06

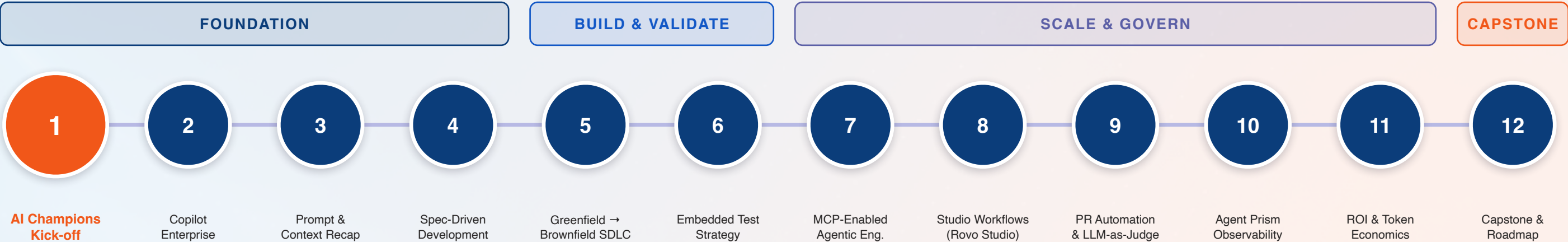
Current → Target Canvas

10 min

Workshop output: workflow canvas + success metrics that feed every later module.

Module 01 in the 12-Module Journey

An incremental engineering journey, not a tour of AI tools — every later module assumes what we establish today.



Why this matters: Copilot capability-building in Module 2, the spec discipline in Module 4, and the brownfield firmware work in Module 5 all assume the shared vocabulary, risk awareness, and success metrics this module establishes.

From Isolated Experimentation to Disciplined Adoption

The shift this programme exists to drive across Process, Industrial & Building Automation engineering.

TODAY — ISOLATED AI EXPERIMENTATION

- Individual engineers try Copilot ad hoc, with no shared context-engineering discipline across PA / IA / BA teams.
- No consistent view of what AI actually changed inside safety- and timing-sensitive C/C++ firmware.
- Inconsistent trust calibration — some engineers over-trust closed-toolchain suggestions, others ignore Copilot entirely.
- Zero specification or traceability discipline: prompt-only development with no link back to requirements.
- No visibility into token cost, usage patterns, or where AI-assisted work is actually paying off.



TARGET — CHAMPION-LED ENTERPRISE ADOPTION

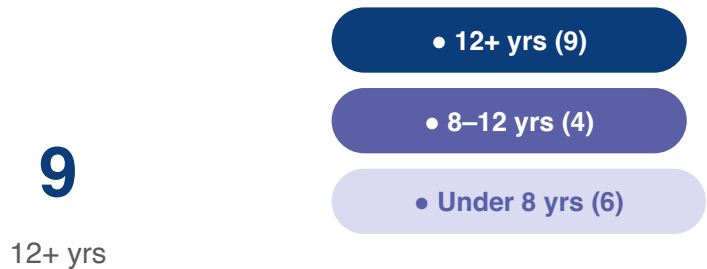
- A designated AI Champion per GBE (PA · IA · BA) anchors practice and answers day-to-day “can we trust this?” questions.
- Shared specification + context-engineering discipline (Modules 3–4) applied consistently to embedded repositories.
- AI usage made visible through Agent Prism — traces, token cost, quality signals, adoption — not a black box.
- Explicit, role-based expectations for Architects, Developers, Testers, DevOps/SRE and Engineering Leads.
- A governed path from greenfield spec-driven builds to safe brownfield firmware evolution.

Who's in the Room: Cohort Composition

28 invited across three Honeywell GBEs — 19 completed the pre-training calibration survey that shaped this module.

Participants by Business Unit & Role Tier

Calibration Survey — Experience Mix (n=19)



9 of 19 respondents rate their C/C++ “Advanced,” another 4 “Expert” — deep skill, not novice curiosity.



11 of 19

TRUST AI ONLY WITH SMALL CHANGES

Only 4 of 19 already trust AI on complex, validated changes; 1 isn't confident at all. Design implication: this programme has to earn broader trust, not assume it.

What the Calibration Survey Told Us

Real answers from 19 Honeywell engineers, collected before this module was finalized — not assumptions.

TOP TRAINING ASKS — PICK UP TO THREE



Signal for the programme: MCP and agentic workflows top the list by a wide margin — confirming Module 7's emphasis before participants ever asked for it.

WHAT NEEDS THE CLOSEST HUMAN CHECK



Real judgment, already present: 18 of 19 correctly said they would NOT approve a PR that compiles and passes tests but uses more memory and breaks an established design pattern.

Defining the AI Champion Persona

Three roles in the enterprise adoption model — today's cohort formally carries the first.



AI Champion

9 in this cohort

Hands-on driver embedded inside real PA/IA/BA firmware work. Pilots Copilot, SDD and MCP workflows on live repositories, surfaces blockers, and sets the local example for “what good looks like.”



AI Evangelist

Emerging from participants

Peer-level influence. Shares wins and lessons from pilots, helps colleagues past first-attempt friction, and feeds practical feedback into the Champion network.



AI Sponsor

Engineering leadership

Removes organisational blockers — tooling access, procurement (Copilot Enterprise seats, Agent Prism), governance sign-off — and protects time for the programme.

Cohort reality: 19 of 28 participants join today as General Participants. Part of this module's job is turning a share of them into tomorrow's Evangelists — not just running Champions through content.

What “Good” Looks Like, By Role

Common foundation for everyone; deeper hands-on emphasis for the technical roles.



Architects

[DEEP-DIVE](#)

Judge where AI-assisted change is safe against firmware architecture, HAL boundaries & interface contracts.



Testers

[DEEP-DIVE](#)

Own the embedded test strategy that proves SDD-developed features preserve behaviour and meet acceptance criteria.



Engineering Leads

[DEEP-DIVE](#)

Set adoption pace, own the ROI/governance conversation, and translate Agent Prism signals into decisions.



Embedded / Firmware Developers

[DEEP-DIVE](#)

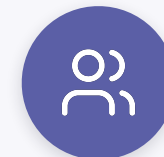
Drive Copilot + SDD end-to-end on real C/C++ repositories — greenfield build and brownfield enhancement.



DevOps / SRE

[DEEP-DIVE](#)

Wire MCP-enabled workflows into build/test/CI safely, with governed tool permissions and failure handling.

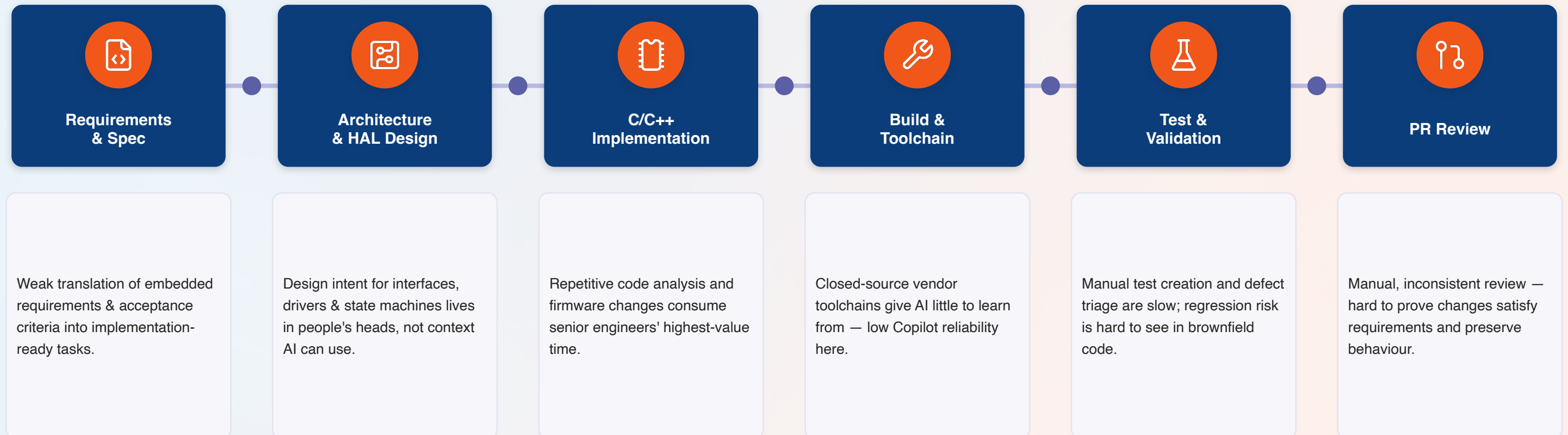


Program / Product Roles

Share the common foundation — enterprise context, risk awareness, success metrics — without the deep technical labs.

The Embedded Delivery Lifecycle — Where It Hurts Today

Every stage participants named in past retros maps to a business problem this programme is built to solve.



The throughline: Every one of these six friction points is a named “business problem this programme solves” — Module 01 just makes the mapping explicit before AI touches any of it.

Risks of Letting AI Loose on Embedded Firmware, Unmanaged

Business & engineering risk this programme's governance model exists to contain.



HIGH

Silent architecture & standards violations

AI-generated C/C++ changes that violate firmware architecture or coding standards without anyone catching it.



HIGH

Hard-to-diagnose brownfield regressions

Broken interfaces or state-machine behaviour in an existing codebase, discovered late — or in the field.



HIGH

Closed-toolchain & RTOS hallucination

Vendor toolchains, RTOS internals & AUTOSAR/CAPL have thin public training data — Copilot confidently guesses wrong.



MEDIUM

Prompt-only, spec-free development

No traceability from requirement to acceptance criteria to implementation — nothing to validate against.



MEDIUM

Context inflation & token waste

Large repos, generated headers & design docs supplied without task-scoped context — cost and noise climb.



MEDIUM

Zero usage & cost visibility

No traces, quality signals or ROI evidence without Agent Prism — adoption decisions made blind.

The cohort already senses this: in the pre-training calibration survey, 18 of 19 engineers correctly declined to approve a PR that compiled and passed tests but broke memory limits and design patterns — this module turns that instinct into policy.

Agent Prism: The Monitoring Lens We'll Use All Week

Introduced today as a concept; hands-on observability arrives in Module 10.



Traces & Replay

Step back through what an agent actually did, in what order.



Failure Patterns

Where and why agent runs go wrong on embedded tasks.



Token Usage & Cost Leakage

Where AI spend is going — and whether it's earning its keep.



Quality & Control Signals

Confidence and quality indicators before code reaches review.



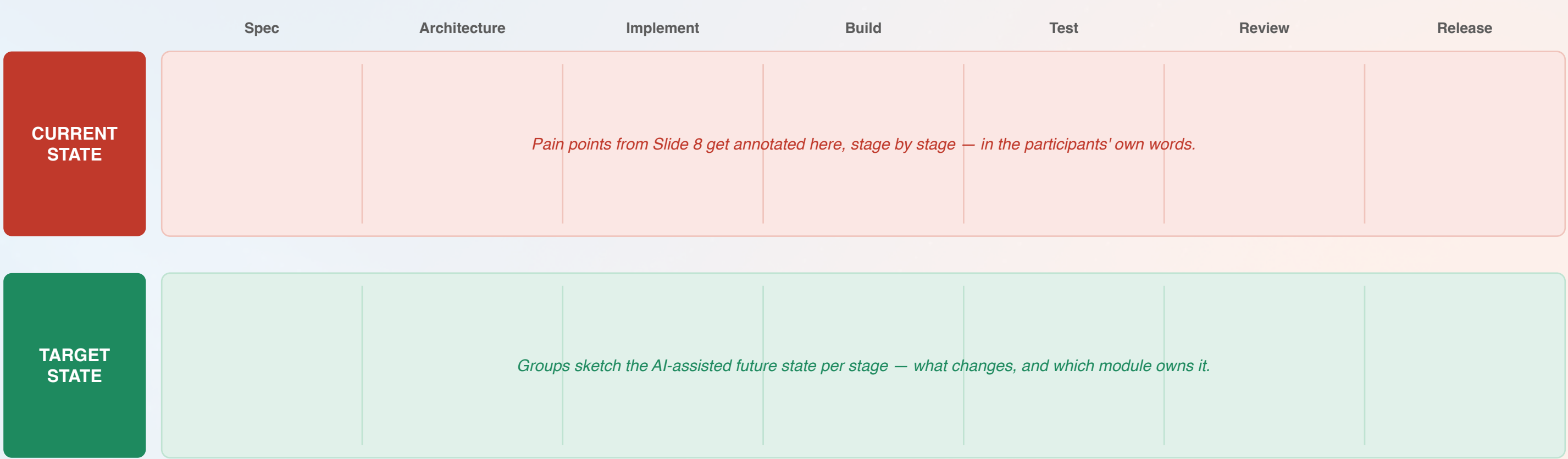
Adoption Tracking

Who is actually using AI-assisted workflows, and how often.

Procurement flag: Agent Prism sandbox access has no public pricing yet. Confirm the procurement path now so it isn't a blocker by Module 10.

Workshop: Mapping Current-State vs Target-State Workflow

10-minute small-group exercise — grouped by GBE, annotating the lifecycle from Slide 8.



Group by GBE

PA · IA · BA teams work their own lifecycle — real repos, real pain.



Split Greenfield / Brownfield

Each group tags stages as new-build vs existing-codebase reality.



Output Feeds Forward

Canvas becomes direct input to the success-metrics worksheet (next slide).

Defining Success Before We Touch a Single Prompt

Workshop output #2: the success-metrics worksheet and the representative use cases Module 5 will build on.

SUCCESS METRICS WORKSHEET



Feature / defect implementation cycle time



PR review latency & review-cycle count



Regression & defect rate in brownfield changes



AI cost per accepted, production-quality change



Adoption rate across PA / IA / BA engineers

REPRESENTATIVE USE CASES — ONE PAIR PER GBE



Process Automation

Greenfield candidate
(new feature, spec-driven)

Brownfield candidate
(existing firmware, safe change)



Industrial Automation

Greenfield candidate
(new feature, spec-driven)

Brownfield candidate
(existing firmware, safe change)



Building Automation

Greenfield candidate
(new feature, spec-driven)

Brownfield candidate
(existing firmware, safe change)

These six candidates carry forward into Module 5's greenfield & brownfield hands-on labs.

Module 01 Outcomes & What's Next

Everything below should be true before we open GitHub Copilot in Module 02.



Enterprise context & business case for AI-assisted embedded engineering established



Role-based expectations set for Architects, Developers, Testers, DevOps/SRE, Leads



Embedded delivery lifecycle pain points mapped stage by stage



Current-state → target-state workflow canvas drafted, by GBE



AI Champion / Evangelist / Sponsor persona model understood, cohort roles named



Business & engineering risks of unmanaged AI usage named and owned



Agent Prism introduced as the monitoring lens for the rest of the programme



Success-metrics worksheet + 6 representative greenfield/brownfield use cases drafted



NEXT — MODULE 02: GITHUB COPILOT ENTERPRISE FEATURES FOR EMBEDDED ENGINEERING

2 hours · Hands-on with chat, inline assist, code explanation, refactoring, debugging, test generation & agent mode on a sample C/C++ embedded repository.

Questions & Discussion

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